

CARALLUMA

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APOCYNACEAE

Genus of perennial succulents with 4–6-ribbed, fingerlike, blue-gray or blue-green to purple stems. Needs sun and extremely well-drained soil. Water sparingly, only in the growing season. May be difficult to grow. Propagate by seed or stem cuttings in summer.

**C. europaea**, syn. *Stapelia europaea*. Clump-forming, perennial succulent. H 8in (20cm), S 3ft (1m). Rough, 4-angled, erect to procumbent, gray stems often arch over and root. Has clusters of small, star-shaped, yellow and brownish-purple flowers near stem crown from mid- to late summer, then twin-horned, gray seed pods. Flowers smell faintly of rotten meat. One of the easier species to grow. Z11–12 (min. 50°F/10°C)

**C. joannis** illus. p.494.

CARDAMINE

Bitter cress

BRASSICACEAE/CRUCIFERAE

Genus of spring-flowering annuals and perennials. Some are weeds, but others are suitable for informal and woodland gardens. Requires sun or semishade and moist soil. Propagate by seed or by division in fall.

**C. enneaphylla**, syn. *Dentaria enneaphylla*. Lax perennial spreading by fleshy, horizontal rootstocks. H 12–24in (30–60cm), S 18–24in (45–60cm). In spring, nodding, pale yellow or white flowers open at the ends of shoots arising from deeply divided leaves. Z7–9

**C. latifolia**. See *C. raphanifolia*.

**C. pentaphylla**, syn. *Dentaria digitata*, *Dentaria pentaphylla*, illus. p.260.

**C. pratensis** (Cuckoo flower, Lady’s smock) illus p.438. **‘Flore Pleno’** is a neat, clump-forming perennial. H 8in (20cm), S 12in (30cm). Divided, mid-green leaves have rounded leaflets. Bears dense sheaves of double, lilac flowers in spring. May be propagated by leaf-tip cuttings in mid-summer. Prefers a moist or wet site. Z4–8

**C. raphanifolia**, syn. *C. latifolia*, illus. p.438.

**C. trifolia** illus. p.348.

CARDIOCRINUM

Giant lily

LILIACEAE

Genus of summer-flowering, lilylike bulbs, grown for their spectacular flowers and attractive, heart-shaped, veined leaves. Needs partial shade and deep, humus-rich, moist soil. Plant bulbs just below soil surface, in fall. Water well in summer and mulch with humus. Provide a deep mulch in winter. After flowering, main bulb dies, but produces offsets. To produce flowers in up to 5 years, propagate by offsets in fall; may also be propagated by seed in fall or winter and will then flower in 7 years.

**C. giganteum** (Giant lily) illus. p.385.

**var. yunnanense** is a sturdy, leafy-stemmed bulb. H 5–6ft (1.5–2m), S 2½–3ft (75–100cm). Has bold, heart-shaped, bronze-green leaves. Produces long spikes of fragrant, pendent, trumpet-shaped, cream flowers, 6in (15cm) long, with

purple-red streaks inside and often tinted green, in summer. Decorative seed heads follow. Z7–9

CARDIOSPERMUM

Heart seed

SAPINDACEAE

Genus of evergreen, perennial, woody-stemmed tendril climbers, grown mainly for their attractive fruits. Useful for covering bushes or trellises. Grow in full light and any soil. Propagate by seed in spring.

**C. halicacabum** (Balloon vine, Heart pea, Love-in-a-puff, Winter cherry). Perennial, tendril climber, usually grown as an annual or biennial. H to 10ft (3m). Has toothed leaves of 2 oblong leaflets. Inconspicuous, whitish flowers are produced in summer, followed by downy, spherical, inflated, 3-angled, straw-colored fruits containing black seeds, each with a heart-shaped, white spot. Z10–12 (min. 45–50°F/7–10°C)

CAREX

Sedge

CYPERACEAE

See also GRASSES, BAMBOOS, RUSHES, and SEDGES.

**C. buchananii** (Leatherleaf sedge). Evergreen, tuft-forming, perennial sedge. H to 2ft (60cm), S 8in (20cm). Very narrow, copper-colored leaves turn red toward base. Solid, triangular stems bear tiny, brown spikelets in summer. Z6–9

**C. elata ‘Aurea’** (Bowles’s golden sedge) illus. p.445.

**C. flagellifera** illus. p.289.

**C. grayi** (Mace sedge). Evergreen, tuft-forming, perennial sedge. H to 2ft (60cm), S 8in (20cm). Has bright green leaves. Large, female spikelets, borne in summer, mature to pointed, knobbly, greenish-brown fruits. Z5–9

**C. hachijoensis ‘Evergold’**. See *C. oshimensis* ‘Evergold’.

**C. morrowii** of gardens. See *C. oshimensis*.

**C. oshimensis**, syn. *C. morrowii* of gardens. Evergreen, tuft-forming, perennial sedge. H 8–20in (20–50cm), S 8–10in (20–25cm). Has narrow, mid-green leaves. Solid, triangular stems bear insignificant spikelets in summer. Z5–9.

**‘Evergold’** (syn. *C. hachijoensis* ‘Evergold’) illus. p.288.

**C. pendula** (Pendulous sedge). Evergreen, tuft-forming, graceful perennial. H 3ft (1m), S 1ft (30cm). Has narrow, green leaves, 18in (45cm) long. Solid, triangular stems freely produce pendent, greenish-brown flower spikes in summer. Z5–9

CARISSA

APOCYNACEAE

Genus of evergreen, spring- to summer-flowering shrubs and small trees, grown for their flowers and overall appearance. Mostly Needs partial shade and well-drained soil. Water containerized specimens moderately, less when temperatures are low. Propagate by seed when ripe or in spring or by semiripe cuttings in summer. ☉ Seeds are poisonous.

**C. grandiflora**. See *C. macrocarpa*.

**C. macrocarpa**, syn. *C. grandiflora* (Natal plum). **‘Tuttlei’** illus. p.453.

**C. spectabilis**. See *Acokanthera oblongifolia*.

CARLINA

ASTERACEAE/COMPOSITAE

Genus of annuals and perennials, grown for their ornamental flower heads, which in most species are good for drying. Requires a sunny position and well-drained soil. Propagate by seed: annuals in spring, perennials in fall.

**C. acaulis** (Alpine thistle) illus. p.361.

CARMICHAELIA

SYN. CHORDOSPARTIUM, NOTOSPARTIUM

PAPILIONACEAE/LEGUMINOSAE

Genus of deciduous shrubs, occasionally small trees, grown for their profusion of tiny flowers in summer. Flattened, green shoots assume function of leaves. Needs full sun and well-drained soil. Cut out dead wood in spring. Propagate by semiripe cuttings in summer or by seed in fall or spring.

**C. carmichaeliae**, syn. *Notospartium carmichaeliae* (Pink broom). Leafless, arching shrub. H 6ft (2m), S 5ft (1.5m). In midsummer, bears short, dense spikes of pealike, purple-blotched, pink flowers on slender, drooping, green shoots. Z8–10

**C. enysii**. Deciduous, mound-forming, dense shrub. H 5 and S 1ft (30cm). Shoots are rigid. Pealike, violet flowers are produced in midsummer. Best grown in a rock garden. Z8–10

**C. stevensonii**, syn. *Chordospartium stevensonii* (Weeping broom). Deciduous, almost leafless, arching shrub or small tree. H 10ft (3m), S 6ft (2m). Produces small, pealike, purplish-pink flowers in cylindrical racemes in midsummer. Z8–10

CARNEGIEA

CACTACEAE

Genus of one species of very slow-growing, perennial cactus with thick, 12–24-ribbed, spiny stems. Unlikely to flower or branch at less than 12ft (4m) high. Requires full sun and very well-drained soil. Propagate by seed in spring or summer.

**C. gigantea** (Saguaro cactus) illus. p.492.

CARPENTERIA

HYDRANGEACEAE

Genus of one species of evergreen shrub, cultivated for its flowers and foliage. Grows well against a warm, sunny wall. Prefers full sun and fairly moist, but well-drained soil. Propagate by seed in fall or by greenwood or semiripe cuttings in summer.

**C. californica** illus. p.197.

CARPINUS

Hornbeam

BETULACEAE

Genus of deciduous trees, grown for their foliage, fall color and clusters of small, winged nuts. Needs sun or semishade and

fertile, well-drained soil. Propagate species by seed in fall, cultivars by budding in late summer.

**C. betulus** (Common hornbeam). Deciduous, round-headed tree. H 80ft (25m), S 70ft (20m). Has a fluted trunk and ovate, prominently veined, dark green leaves that turn yellow and orange in fall. Bears green catkins from late spring to fall, when clusters of winged nuts are borne. Z5–7. **‘Fastigiata’** (syn. *C.b. ‘Pyramidalis’*) illus. p.93. **‘Pyramidalis’** see *C.b. ‘Fastigiata’*.

**C. caroliniana** (American hornbeam). Deciduous, spreading tree with branches that droop at tips. H and S 30ft (10m). Has a fluted, gray trunk, green catkins in spring, and ovate, bright green leaves that turn orange and red in fall, when clusters of winged nuts are borne. Z5–7

**C. turczaninowii**. Deciduous, spreading tree of graceful habit. H 40ft (12m), S 30ft (10m). Green catkins are borne in spring. Produces clusters of small, winged nuts in fall, when the small, ovate, glossy, deep green leaves turn orange. Z5–7

CARPOBROTUS

AIZOACEAE

Genus of mat-forming, perennial succulents with triangular, fleshy, dark green leaves and daisylike flowers. Excellent for binding sandy soils. Needs full sun and well-drained soil. Propagate by seed or by stem cuttings in spring or summer.

**C. edulis** (Hottentot fig, Kaffir fig). Carpeting, perennial succulent. H 6in (15cm), S indefinite. Prostrate, rooting branches bear leaves ⅝in (1.5cm) thick and 5in (12cm) long. Yellow flowers, 5in (12cm) across, open in spring–summer from about noon in sun. Bears edible, figlike, brownish fruits in late summer and fall. Z10–11 (min. 41°F/5°C)

CARTHAMUS

ASTERACEAE/COMPOSITEAE

Genus of upright, hairy annuals and herbaceous perennials. *C. tinctorius* has been used for centuries as a source of red and yellow dye. Grow in full sun in any light, well-drained soil. Flower heads are good for dried arrangements. Excellent in a border or herb garden. Propagate by seed in spring.

**C. tinctorius** (False saffron) illus. p.323.

CARYA

Hickory

JUGLANDACEAE

Genus of deciduous trees, grown for their stately habit, divided leaves, fall color and, in some cases, edible nuts. Has insignificant flowers in spring. Requires sun or semishade and deep, fertile soil. Plant young seedlings in a permanent position during their first year since older plants resent transplanting. Propagate by seed in fall.

**C. cordiformis** (Bitternut, Bitternut hickory). Vigorous, deciduous, spreading tree. H 80ft (25m), S 50ft (15m). Bark is smooth at first, later fissured. Bright yellow, winter leaf buds develop into large, dark green leaves, with 5 to 9 ovate to oblong leaflets; these turn yellow in fall.